

Areas of Interest

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1.1. Air Domain Technologies

- 1.1.1. Human Performance Optimization
- 1.1.2. Aerospace Medicine and Operational Health
- 1.1.3. Warfighter Survivability and Protection
- 1.1.4. Cognitive and Behavioral Sciences
- 1.1.5. Advanced Air Vehicles
- 1.1.6. Air Propulsion Systems
- 1.1.7. Aerodynamics and Flight Sciences
- 1.1.8. Hypersonics
- 1.1.9. Advanced Weapons Technologies
- 1.1.10. Energetics and Propellants
- 1.1.11. Guidance Navigation and Control
- 1.1.12. Munitions Survivability and Countermeasures

1.2. Space Domain Technologies

- 1.2.1. Space Access and Orbital Systems
- 1.2.2. Spacecraft and Satellite Technologies
- 1.2.3. Space Power and Propulsion
- 1.2.4. On-Orbit Operations and Autonomy
- 1.2.5. Space Situational Awareness
- 1.2.6. Resilient and Survivable Space Architecture
- 1.2.7. Space Environment and Effects
- 1.2.8. Position, Navigation, and Timing (PNT) from and for Space

1.3. Cyberspace/Electronic Warfare Domain Technologies

- 1.3.1. Cyber Operations and Cybersecurity
- 1.3.2. Communications and Networks
- 1.3.3. Electronic Warfare and Spectrum Operations
- 1.3.4. Quantum Technologies
- 1.3.5. Electro-Optical and Infrared Sensors
- 1.3.6. Radar and Radio Frequency Sensors
- 1.3.7. Multispectral and Hyperspectral Sensing
- 1.3.8. Signal Processing and Sensor Fusion
- 1.3.9. Quantum Sensing and Advanced Properties
- 1.3.10. Directed Energy Sensing and Countermeasures
- 1.3.11. High-Energy Laser Systems
- 1.3.12. Photonics
- 1.3.13. High-Power Microwave Technologies
- 1.3.14. Beam Control and Adaptive Optics
- 1.3.15. Directed Energy Effects and Countermeasures

45 1.3.16. Cybersecurity for Digital Environments

46 1.3.17. Navigation Warfare and Counter PNT

47
48 1.4. Cross-Domain Technologies

49 1.4.1. Human Systems Integration

50 1.4.2. Human-Machine Teaming and Interfaces

51 1.4.3. Training and Simulation Technologies

52 1.4.4. Rapid Multi-Domain Integration

53 1.4.5. Advanced Systems Engineering

54 1.4.6. Integrated Vehicle Health Management

55 1.4.7. Power and Thermal Management

56 1.4.8. Advanced Materials Development

57 1.4.9. Additive Manufacturing

58 1.4.10. Materials Processing and Fabrication

59 1.4.11. Materials Characterization and Testing

60 1.4.12. Environmentally Responsive and Smart Materials

61 1.4.13. Microelectronics

62 1.4.14. Model-Based Systems Engineering, Digital Engineering, and Digital Thread

63 1.4.15. Advanced Simulation and Virtual Prototyping

64 1.4.16. Data Analytics and Artificial Intelligence

65 1.4.17. Additive Manufacturing Digital Integration

66 1.4.18. Low Cost, Size, Weight, and Performance (C-SWAP) Alternative PNT
67 Systems for Attritable Systems

68 1.4.19. Modular Open Architecture approaches to Resilient PNT

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70 **2. Definitions**

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72 2.1. Air Domain Technologies

73 2.1.1. Human Performance Optimization - Enhancing human capabilities and
74 resilience across diverse operational environments through the application
75 of scientific principles, advanced technologies, and personalized
76 interventions. This includes, but is not limited to, physical conditioning,
77 cognitive enhancement, stress management, and optimized human-
78 machine interfaces techniques.

79 2.1.2. Aerospace Medicine and Operational Health - Advancing medical research,
80 developing countermeasures, and implementing strategies to optimize the
81 health, safety, and performance of personnel in demanding operational
82 environments. This includes physiological monitoring, environmental
83 protection, human performance enhancement, and medical support for
84 air, space, and cyber operations.

85 2.1.3. Warfighter Survivability and Protection - Developing and integrating
86 advanced technologies and strategies to protect personnel from a wide
87 range of threats and hazards in operational environments. This includes

- 88 personal protective equipment, threat detection systems, casualty care,
89 and environmental control measures applicable across various domains.
- 90 2.1.4. Cognitive and Behavioral Sciences - Understanding and leveraging human
91 cognitive and behavioral principles to improve decision-making, enhance
92 performance, and optimize human-system interaction in complex and
93 dynamic environments. This encompasses areas such as perception,
94 attention, memory, reasoning, and social interaction.
- 95 2.1.5. Advanced Air Vehicles - Designing and developing next-generation aircraft
96 technologies for diverse operational roles, including high-speed flight,
97 enhanced maneuverability, stealth capabilities, and optionally manned
98 configurations to extend operational reach and survivability.
- 99 2.1.6. Air Propulsion Systems - Innovating propulsion technologies such as
100 turbine engines, hypersonic propulsion (scramjets, ramjets), and advanced
101 powerplants to improve efficiency, speed, and performance of aerospace
102 platforms
- 103 2.1.7. Aerodynamics and Flight Sciences - Conducting fundamental research in
104 aerodynamics, flight control, stability, and aircraft performance to enable
105 improved maneuverability, endurance, and stealth characteristics.
- 106 2.1.8. Hypersonics - Developing technologies for hypersonic flight—speeds
107 greater than Mach 5—including materials, propulsion, guidance, and
108 control systems to enable rapid global strike and ISR capabilities.
- 109 2.1.9. Advanced Weapons Technologies - Developing innovative kinetic and non-
110 kinetic weapons, including smart munitions, precision-guided munitions,
111 and novel warhead designs to increase effectiveness against evolving
112 threats.
- 113 2.1.10. Energetics and Propellants - Researching advanced energetic materials,
114 propellants, and explosives that provide greater performance, safety, and
115 reduced environmental impact.
- 116 2.1.11. Guidance Navigation and Control - Developing advanced guidance and
117 control systems for improved accuracy, maneuverability, and target
118 discrimination in complex operational environments.
- 119 2.1.12. Munitions Survivability and Countermeasures - Enhancing munition
120 robustness against countermeasures and hostile environments to ensure
121 reliable performance under combat conditions.
- 122
- 123 2.2. Space Domain Technologies
- 124 2.2.1. Space Access and Orbital Systems - Supporting the development of launch
125 vehicles, space propulsion, and space maneuvering systems to ensure
126 reliable, responsive, and cost-effective access to space.
- 127 2.2.2. Spacecraft and Satellite Technologies - Developing advanced technologies
128 and innovative architectures for space-based assets to enhance resilience,
129 improve performance, and expand mission capabilities. This encompasses
130 satellite bus design, payload development, communication systems, on-
131 orbit servicing, and other critical technologies for space operations.

- 2.2.3. Space Power and Propulsion - Innovating power generation (solar arrays, energy storage) and electric propulsion systems to extend satellite life and improve maneuverability and station-keeping.
- 2.2.4. On-Orbit Operations and Autonomy - Advancing technologies for autonomous spacecraft operations, rendezvous and proximity operations, on-orbit servicing, and debris mitigation to maintain space asset functionality.
- 2.2.5. Space Situational Awareness - Enhancing sensors, data processing, and analytics to detect, track, and characterize objects and threats in the space environment, supporting space domain awareness.
- 2.2.6. Resilient and Survivable Space Architecture - Researching distributed satellite constellations, hardened systems, and defensive measures to ensure continuity of space operations in contested or degraded environments.
- 2.2.7. Space Environment and Effects - Studying space weather, radiation effects, and the orbital environment to develop mitigation strategies that protect space assets and maintain operational readiness.
- 2.2.8. Position, Navigation, and Timing (PNT) from and for Space - The development, deployment, and exploitation of space-based systems and technologies that enable precise geolocation, accurate time dissemination, and resilient navigation capabilities for terrestrial, airborne, maritime, and spaceborne users.
- 2.3. Cyberspace/Electronic Warfare Domain Technologies
 - 2.3.1. Cyber Operations and Cybersecurity - Developing and implementing comprehensive strategies, technologies, and practices to secure cyberspace, protect critical assets, and enable effective cyber operations in support of national security objectives. This encompasses offensive and defensive cyber capabilities, threat intelligence, vulnerability management, and incident response.
 - 2.3.2. Communications and Networks - Innovating next-generation, resilient, and secure communications technologies, including software-defined radios, dynamic spectrum management, and high-capacity tactical networks supporting contested environments.
 - 2.3.3. Electronic Warfare and Spectrum Operations - Researching techniques for electromagnetic spectrum dominance, including signal processing, electronic attack and protection, and spectrum sensing.
 - 2.3.4. Quantum Technologies - Advanced systems that leverage the principles of quantum mechanics—such as superposition and entanglement—to enable next-generation computing, communication, and sensing capabilities.
 - 2.3.5. Electro-Optical and Infrared Sensors - Developing advanced imaging and sensing systems operating across visible, infrared, and ultraviolet spectra

175 for target detection, tracking, and identification in all-weather and
176 contested environments.

177 2.3.6. Radar and Radio Frequency Sensors - Innovating radar technologies
178 including active electronically scanned arrays (AESAs), synthetic aperture
179 radar (SAR), and low-probability-of-intercept/low-probability-of-detection
180 (LPI/LPD) radars to enhance detection and tracking of air, surface, and
181 maritime targets.

182 2.3.7. Multispectral and Hyperspectral Sensing - Creating sensors capable of
183 collecting data across multiple spectral bands for improved material
184 discrimination, camouflage detection, and environmental monitoring.

185 2.3.8. Signal Processing and Sensor Fusion - Advancing algorithms and
186 architectures to integrate data from multiple sensor types for enhanced
187 target recognition, situational awareness, and reduced false alarms.

188 2.3.9. Quantum Sensing and Advanced Properties - Exploring emerging quantum
189 technologies and photonic devices to achieve breakthroughs in sensitivity,
190 resolution, and sensor miniaturization.

191 2.3.10. Directed Energy Sensing and Countermeasures - Developing sensors that
192 detect and characterize directed energy threats and support electronic
193 warfare and self-protection systems.

194 2.3.11. High-Energy Laser Systems - Developing scalable, high-power laser
195 technologies for precision engagement of air, ground, and space targets.
196 This includes solid-state lasers, fiber lasers, and beam control systems for
197 weaponization.

198 2.3.12. High-Power Microwave Technologies - Researching microwave energy
199 sources and systems to disable, disrupt, or destroy enemy electronic
200 systems, communications, and sensors without kinetic effects.

201 2.3.13. Beam Control and Adaptive Optics - Innovating advanced beam steering,
202 adaptive optics, and atmospheric compensation techniques to maintain
203 laser beam quality and accuracy over long distances and through turbulent
204 environments.

205 2.3.14. Directed Energy Effects and Countermeasures - Studying the physical and
206 operational effects of directed energy weapons on targets and developing
207 defensive measures against hostile directed energy threats.

208 2.3.15. Cybersecurity for Digital Environments - Protecting digital assets,
209 infrastructure, and information from cyber threats through the
210 development and implementation of advanced security measures and
211 resilient architectures. This encompasses threat detection, vulnerability
212 assessment, intrusion prevention, and incident response to ensure the
213 confidentiality, integrity, and availability of critical systems.

214 2.3.16. Navigation Warfare and Counter PNT - Research and development efforts
215 focused on both disrupting adversary PNT systems and protecting friendly
216 forces from similar attacks, ensuring mission success in GPS-denied or
217 degraded environments. This involves developing both offensive and
218 defensive technologies and strategies related to PNT.

2.4. Cross-Domain Technologies

- 2.4.1. Human Systems Integration - Integrating human capabilities, limitations, and preferences into the design, development, and evaluation of systems to optimize human-machine interaction, improve usability, and enhance overall system effectiveness. This encompasses human factors engineering, user interface design, workload assessment, and cognitive support to ensure safe, efficient, and intuitive operation.
- 2.4.2. Human-Machine Teaming and Interfaces - Developing technologies to improve interaction between humans and AI-enabled systems, optimizing performance across diverse operational environments and enhancing decision-making, mission planning, and situational awareness.
- 2.4.3. Training and Simulation Technologies - Developing and deploying innovative training methodologies and simulation tools to enhance individual and team proficiency, improve operational readiness, and facilitate effective mission rehearsal across a spectrum of operational scenarios. This includes virtual reality, augmented reality, serious gaming, and adaptive learning systems.
- 2.4.4. Rapid Multi-Domain Integration - Combining air, space, cyber, and ground technologies to develop interoperable, networked capabilities that enhance joint and coalition operational effectiveness.
- 2.4.5. Advanced Systems Engineering - Applying holistic and innovative systems engineering approaches to design, develop, integrate, and test complex systems and architectures, ensuring seamless interoperability, optimal performance, and cost-effective solutions across the entire lifecycle. This encompasses requirements engineering, architecture development, modeling and simulation, and verification and validation.
- 2.4.6. Integrated Vehicle Health Management - Creating systems for real-time monitoring and diagnostics of aerospace platforms to enhance safety, readiness, and maintainability.
- 2.4.7. Power and Thermal Management - Advancing technologies for efficient power generation, storage, distribution, and thermal dissipation across multiple platforms and operational scenarios. This encompasses innovations in energy harvesting, power electronics, battery technology, advanced cooling systems, and thermal interface materials to enable high-performance systems in demanding environments.
- 2.4.8. Advanced Materials Development - Creating novel materials with enhanced properties tailored for diverse applications, including but not limited to aerospace structures, electronic components, and protective coatings. These materials may include high-temperature ceramics, lightweight composites, metamaterials, multifunctional materials, and bio-inspired materials. Focus is on improving performance characteristics, durability, and resilience in extreme and contested environments.

- 2.4.9. Additive Manufacturing - Innovating and maturing additive manufacturing techniques for on-demand production of complex geometries, customized parts, and multi-material components. This encompasses advancements in materials, processes, modeling, and quality control to enable rapid prototyping, distributed manufacturing, and agile sustainment solutions across various domains.
- 2.4.10. Materials Processing and Fabrication - Developing and optimizing advanced processing and fabrication techniques for creating materials and components with tailored properties and microstructures. This includes methods such as nanomanufacturing, advanced joining, surface engineering, and coatings to enhance performance, reliability, and longevity in a wide range of applications.
- 2.4.11. Materials Characterization and Testing - Using experimental and computational techniques to understand the structure, properties, and performance of materials under relevant operational conditions, enabling the development of advanced aerospace systems.
- 2.4.12. Environmentally Responsive and Smart Materials - Developing materials that adapt to their environment or provide sensing and self-healing capabilities to enhance system resilience and functionality.
- 2.4.13. Model-Based Systems Engineering, Digital Engineering and Digital Thread, Advanced Simulation and Virtual Prototyping - Creating and utilizing comprehensive digital representations of systems and their lifecycles to enable improved design, analysis, verification, and validation. This encompasses digital twins, virtual prototypes, and simulation environments to optimize performance, reduce costs, and accelerate development timelines.
- 2.4.14. Data Analytics and Artificial Intelligence - Developing and applying advanced algorithms, machine learning techniques, and data-driven approaches to extract actionable insights from complex datasets and enable intelligent decision-making. This encompasses predictive analytics, pattern recognition, anomaly detection, and automated reasoning to improve situational awareness, optimize operations, and enhance system performance.
- 2.4.15. Additive Manufacturing Digital Integration - Integrating additive manufacturing processes with digital design, simulation, and analysis tools to enable closed-loop optimization, enhanced design freedom, and improved part quality. This encompasses generative design, topology optimization, process simulation, and in-situ monitoring to maximize the benefits of additive manufacturing.
- 2.4.16. Low C-SWAP Alternative PNT Systems for Attritable Systems – Developing affordable and expendable PNT solutions to ensure resilient navigation and timing capabilities for unmanned, potentially disposable aircraft in GPS-denied environments prioritizing C-SWAP requirements

305 2.4.17. Modular Open Architecture approaches to Resilient PNT - Developing
306 flexible and adaptable PNT systems that use standardized interfaces,
307 enabling rapid integration of new technologies and algorithms to maintain
308 reliable navigation and timing in contested environments. This modularity
309 facilitates quicker upgrades and responses to emerging threats against
310 GPS.